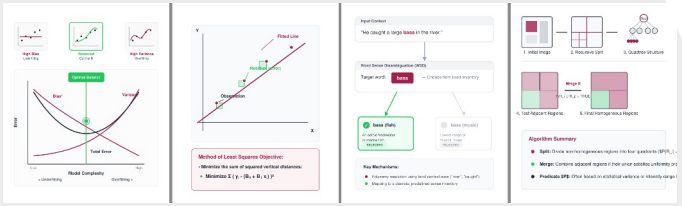
IPD PROJECT REVIEW 2026

TITLE PAGE

- **Project – Learnova**
- **Theme – EdTech · Generative AI**
- **Category – Software**
- **Group No. – 1 (AI & DS, DJSCE)**
- **Team – Swayam · Kawaljeet · Jeevesh · Rishit**
- **HoD – Dr. Pratik Kanani**



Learnova

LEARNOVA — IDEA

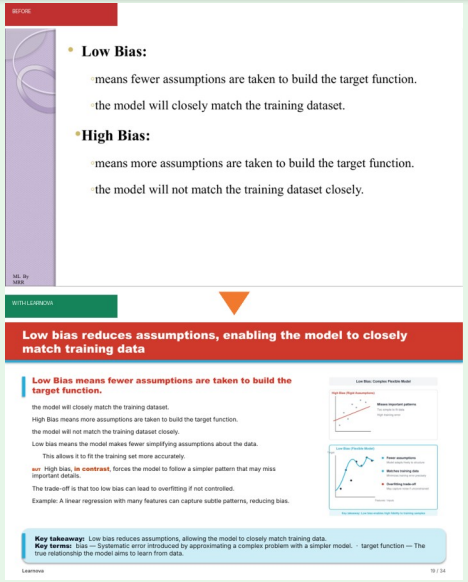
5 real lecture decks: 83 untitled slides · 0 diagrams · 0 quizzes

Proposed Solution: Teachers reuse old decks for years and students revise from them before every exam, yet most are walls of bullets with pictures of equations. Learnova takes the deck a teacher already has (PPTX, PDF, Word notes or a topic) and rebuilds it as an editable PowerPoint, an interactive web deck and a study pack — every point kept, plus a visual, an illustration, quizzes, solved steps and labs.

Detailed explanation of the proposed solution

- **Slide Reader:** every page read with titles, bullet levels, tables, figures; OCR only for picture slides.
- **Decision Engine:** 27 detectors pick process, timeline, comparison, definition, chart or cards — with a reason.
- **Assertion Titles:** each title states the point of its slide; split parts are named, never "(2/5)".
- **Illustrations:** AI-drawn diagram, teacher's figure or offline scene on every slide, linked as an HTML page.
- **Explainers & Labs:** beginner step-by-step solutions and hands-on labs for hard topics.
- **Outputs:** editable PPTX, web deck, study pack, share link, tutor.

How it addresses the problem



Innovation and uniqueness of the solution

- **Redesign, not generate:** starts from the teacher's own deck; other tools start from a prompt.
- **Faithful by design:** source text kept verbatim; 98–100% of facts retained on real decks.
- **Explainable choices:** every visual decision shows its rule and confidence.
- **PSF metric + CLASS:** our own slide-fitness score and load-optimal slide splitting.
- **Two safety nets:** 5 Gemini keys rotate; if all fail, offline rules finish the deck.
- **Verified AI output:** diagrams sanitised, labs code-checked, titles grounded in the slide.
- **Works offline:** no paid licence, runs on a college laptop.

PHASE 1 · Sep–Dec 2025

Research and design

- Literature + competitor study
- Problem, architecture, UI plan

PHASE 2 · Jan–Jul 2026

Build the platform

- Prototype V1 → Learnova V2
- Engine, builders, accounts

PHASE 3 · Aug–Oct 2026

Real decks, AI, research

- Reader + OCR, illustrations, labs
- AI key pool, papers, review

LIVE CODE

github.com/SDP42/LearnovaV2

Open source · 774 tests · demo decks

Learnova

TECHNICAL APPROACH

System architecture and technology stack

INPUTS

PowerPoint .pptx
python-pptx · LibreOffice
teacher's old deck, any template

Slide PDF
PyMuPDF
exported or scanned slides

Word / notes
python-docx · Markdown
.docx, markdown, plain text

Typed topic
React form → REST
a syllabus line or prompt

Teacher labs
HTML · JavaScript
own labs added via the API

INGEST

Convert
LibreOffice headless
PPTX → PDF, cached by hash

Slide reader
PyMuPDF geometry rules
titles, levels, tables, figures

OCR decision
Tesseract OCR
only when γ ≤ 0.42, conf ≥ 62

Repair rules
Python · regex · DP
logos, page numbers, glued titles

Colour DNA
Pillow colour clustering
source palette → deck theme

ENGINES

Planner + CLASS
Python dynamic programming
load-optimal split, agenda

Decision engine
27 rule detectors
visual family + reason

Layout engine
glyph metrics · python-pptx
11 pt floor, no broken words

Explainer solvers
SymPy · exact fractions
steps with reasons

PSF scorer + audit
NumPy · pytest
fitness, overflow, facts kept

AI SERVICE

LLM router
Gemini ×5 · Groq · NVIDIA
failover, per-key pacing

AI polish
Gemini 3.5 Flash-Lite
assertion titles, explanations

Topic diagrams
SVG → CairoSVG → PNG
sanitised, linked HTML page

AI labs
JavaScript · Node check
no network, no storage

Offline fallback
rules · scenes · labs
works with no key at all

OUTPUTS

Editable PPTX
python-pptx native shapes
linked pictures, notes

Web deck
Reveal.js · HTML5
reveals, labs, flip cards

Study pack
HTML · CSV (Anki)
flashcards, quiz bank

Share + present
React 18 · FastAPI
view-only link, presenter

Deck tutor
router + deck context
answers only from the deck

Technologies to be used

React · Vite · Tailwind · Reveal.js · Python · FastAPI · SQLite · PyMuPDF
LibreOffice · Tesseract · Gemini · Groq · SymPy · pytest · GitHub · Docker

Frontend: React 18 · Vite · Tailwind CSS · Reveal.js web deck · PWA

Backend: Python 3.12 · FastAPI · Pydantic · job queue · SQLite · OTP auth

Documents: python-pptx · PyMuPDF · LibreOffice · python-docx · Tesseract OCR

AI / NLP: Gemini (5-key pool) · Groq · NVIDIA NIM via failover router

Maths & labs: SymPy exact solvers · JavaScript lab engine · SVG

Quality & DevOps: pytest (774 tests) · layout audit · GitHub · Docker

Data and storage

SQLite for users, decks and jobs; content-hash caches for PDFs, diagrams and labs

API and jobs

FastAPI with about 70 REST routes, background job queue, live progress, OpenAPI docs

Security

OTP e-mail sign-in, hashed codes, rate limits, SVG sanitiser, lab-code gate

Deploy and test

One laptop or server; Vercel-ready front end; 774 pytest tests on GitHub

WORKING PROTOTYPE · real screens from Learnova



Learnova

METHODOLOGY

Six stages from the teacher's old deck to a checked, interactive one

Methodology and process for implementation

1 · READ

- PPTX → PDF with LibreOffice, cached by file hash
- Words read with position, size, weight (PyMuPDF)
- Title rule: size ≥ 1.18 × body size
- OCR only if γ ≤ 0.42, confidence ≥ 62
- Tables and figures kept in place

2 · CLEAN

- Text on ≥ max(3, 0.3n) pages removed: logos, footers
- Glued titles split by dynamic programming
- Split words and wrapped lines joined
- Chart debris and page counters dropped
- Repeated pictures marked as decoration

3 · PLAN

- Pages grouped into topics and sections
- CLASS splits long sections at least load cost
- Agenda, glossary and summary slides added
- Parts get names, never "(2/5)"
- Each part stays under the load limit

4 · DECIDE

- 27 detectors score every slide
- Visual family d* = highest confidence
- Exclusion rules stop clashing choices
- Reason and confidence stored per slide
- Flowchart, algorithm, code, equations, charts…

5 · ENRICH

- AI titles kept only if grounded in the slide
- Topic diagrams from the 5-key Gemini pool
- Exact step-by-step explainers (SymPy)
- Labs: built-in, AI-written + checked, or offline
- Quizzes and flashcards from slide facts

6 · BUILD + AUDIT

- Glyph-measured fitting, 11 pt floor
- Editable PPTX, web deck and study pack
- PSF score, overflow and fact audit
- Caches make every re-run instant
- Notes and alt text on every slide

Our own formulas behind each decision

OCR decision
$\gamma = |I \setminus M| / |I|$ use OCR only if $\gamma \le 0.42$
share of the page that is picture, not text

Slide fitness (PSF)
$PSF = E^{0.40} \cdot (1 - L)^{0.42} \cdot C^{0.18}$
efficiency, cognitive load L, coherence

CLASS splitting
$F(i) = \min_p [F(p) + cost(p, i)]$
$cost = L + 0.9(1 - E) + 0.25 \cdot overflow^2 + 0.35$

Word-safe layout
$\sigma = \max(0.62, \min(1, W / w))$, $s^* \ge 11$ pt
largest font on a 0.5 pt grid that fits

How we built it

Research + design Sep–Dec 2025 → **Build platform** Jan–Jul 2026 → **Test + publish** Aug–Oct 2026

- ✓ **Agile:** weekly sprints, reviewed with our guide every week
- ✓ **Test-first:** 774 automated tests, gold sets of 101 slides
- ✓ **Measured:** every claim on these slides comes from a script
- ✓ **Documented:** IPD diary, report and two research papers

Learnova

FEASIBILITY AND VIABILITY

Running today · pilot-ready · every risk has a safeguard

Analysis of the feasibility of the idea

TECHNICAL — 774 tests
Built and running end to end: FastAPI back end, React front end, 16 pipeline stages as separate modules, 774 automated tests and a layout audit.

DATA — 7 real decks
No dataset to collect: input is the college's own PPTX, PDF and DOCX material. Verified on 7 real decks (15–216 pages) and 101 gold-labelled slides.

ECONOMIC — ₹0 offline
₹0 to run offline; free-tier Gemini keys (pool of 5) are optional. One department PC replaces paid AI-slide plans ($10–20 per teacher per month).

OPERATIONAL — editable
No retraining for staff: upload, review, download. Every slide stays editable, every choice shows its reason; students need only a browser or phone.

SECURITY — sanitised
OTP sign-in with hashed codes and rate limits; AI diagrams sanitised, AI lab code checked before use; share links view-only; data stays on campus.

SCALE — ~35 s / 216 pp
Time grows linearly: a 216-page PDF becomes a full deck in ~35 s offline. Parallel workers, job queue and content-hash caches make re-runs instant.

Potential challenges and risks



7 risks found · 7 safeguards built

We listed what could stop Learnova in a real classroom and built one safeguard for each. Each risk is rated by how likely it is and how much harm it would do if it happened; the right-hand column shows the safeguard already in the code. Residual risk after safeguards: LOW.

Risk	Likely	Impact	Safeguard in code
Messy or scanned PDFs	High	High	Geometry rules + OCR test
AI makes up facts	Med	High	Grounding check, SymPy maths
Keys or quota run out	High	Med	5-key pool + offline rules
Teachers distrust AI	Med	High	Reason + confidence shown
Text overflows slides	Med	Med	Glyph fit, CLASS split
Slow on long decks	Low	Med	Parallel + caches
Unsafe generated code	Low	High	Sanitiser + node --check

Strategies for overcoming these challenges

1. **Faithful Reader** — vs Messy PDFs
Geometry rules, a DP title splitter and the γ ≤ 0.42 OCR test keep messy PDFs, logos and picture slides correct; 98–100% of facts kept.

2. **Grounded AI** — vs Hallucination
An AI title or explanation is used only if it reuses the slide's own words and fits length rules; every number in a solution comes from SymPy.

3. **Key Pool + Offline** — vs Quota / cost
Offline first: free Gemini and Groq key pools are used only when the rule result is weak; spent keys are skipped; offline rules always finish.

4. **Glass-box Decisions** — vs Black-box
Each visual records the detector that fired, its confidence and the runner-up; the teacher can switch it in the editor in one click.

5. **Measured Layout** — vs Broken layout
Text is measured glyph by glyph with an 11 pt floor; long sections split by CLASS. Result on 7 decks: 0 overflow, 0 broken words.

6. **Fast by Design** — vs Slow decks
Diagrams and labs run in parallel (2 requests per key) before the build; with 5 keys 40 diagrams take ≈ 42 s instead of 3 min.

7. **Safe Generated Code** — vs Unsafe code
SVG whitelist removes scripts and links; lab JavaScript must pass node --check and may not use network, eval, cookies or storage.

VERIFY IT YOURSELF scan / click

Code · Tests · Gold sets · Papers

ADOPTION PATH · low-risk, one step at a time

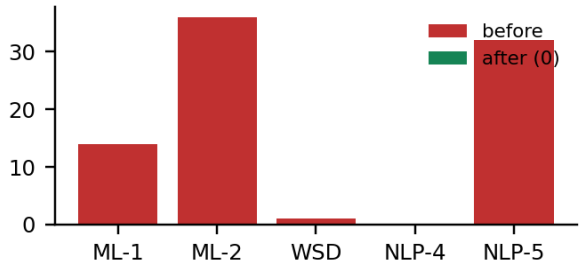
1 · PILOT	2 · DEPT	3 · COLLEGE
Our AI & DS courses; 3–4 faculty	Shared server, deck + lab library	LMS export, regional languages

Learnova

IMPACT AND BENEFITS

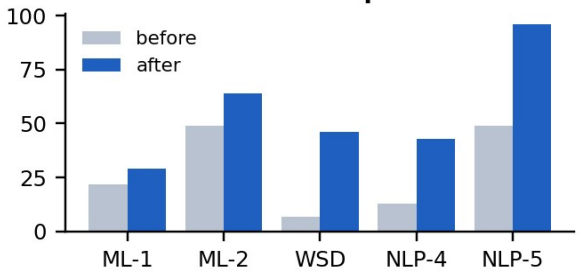
Measured on our own test decks — not on promises

Untitled slides: before → 0



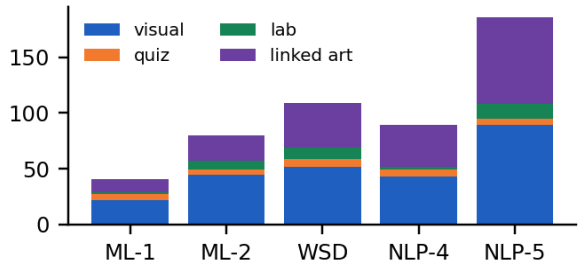
Untitled slides across 5 real decks fell from 83 to 0. Every slide now states its point in a full-sentence title.

Slides with a picture



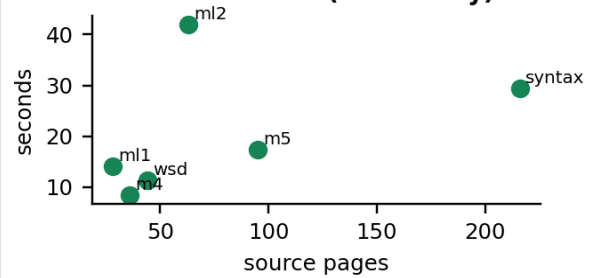
Slides with a picture rose on every deck: the teacher's own figures are kept and AI diagrams are added on topic slides.

Added by Learnova (all 0 before)



Visual slides, quizzes, labs and linked illustrations went from zero to hundreds, all built from the deck's own content.

Time to build (no API key)



Build time grows in a straight line with length: a 216-page PDF becomes a full deck in about 35 s with no API key.

EDUCATIONAL

One idea per slide with a picture, so the eye has one thing to read. Solved problems become step-by-step explainers with a reason for each step, and hard topics get labs and quizzes to practise on.

SOCIAL

Runs offline and free on a college laptop, so students without paid tools, fast internet or a laptop of their own get the same material through a share link on any phone. Plain words and alt text help first-generation learners.

INSTITUTIONAL

Course decks become consistent and accessible across the department. A shared library of decks and labs builds up every semester and supports flipped, remote and revision classes.

TECHNOLOGICAL

Explainable rules instead of a black box, our own PSF score and CLASS splitting, and AI used only where its output can be checked: sanitised diagrams, checked code, grounded titles.

Students
revise with visuals, quizzes and labs

Faculty
old deck redesigned in about a minute

HoD / Department
consistent, reusable course material

Exam preparation
flashcards, quiz bank, cheat sheet

Remote learners
share link on any phone, works offline

Accessibility
readable text, alt text, plain words

83 → 0	251	190	98–100%	0	26 / 26
untitled slides	visual slides added	linked illustrations	facts kept	overflow or broken words	explainers correct

Learnova

LITERATURE REVIEW

What earlier work found, what it left open, and how Learnova closes the gap

Paper	Method	Finding	Gap left open	How Learnova addresses it
Fu et al., 2022 DOC2PPT — AAAI	Neural model turns a research paper into new slides section by section	Readable slides from long documents	Writes new text; drops points; needs training data; not for existing decks	**Starts from the teacher's own deck and keeps every point verbatim**
Sun et al., 2021 D2S — NAACL	Query-based summarisation: a slide title is used to retrieve and write its bullets	Titles can steer content selection	Abstractive text can drift from the source; no visual or layout choice	**Titles are grounded in the slide's words; 27 detectors choose a visual**
Hu & Wan, 2015 PPSGen — IEEE TKDE	Learns sentence importance and fills slides by integer programming	Extractive selection keeps facts	Bullet text only: no diagrams, no layout fitting, no interactivity	**Adds diagrams, labs and quizzes; fits every slide by glyph measurement**
Maynez et al., 2020 Faithfulness — ACL	Human study of hallucination in neural abstractive summaries	Many model summaries contain unsupported content	Generation without checks is unsafe for study material	**AI output is accepted only after grounding, sanitising or code checks**
Garner & Alley, 2013 Assertion–evidence — IJEE	Classroom study comparing sentence-headline slides with topic-bullet slides	Assertion titles + visual evidence improve understanding	Redesigning decks by hand takes teachers many hours	**Writes assertion titles and visuals automatically, in about a minute**
Mackinlay, 1986 APT — ACM TOG	Rules of expressiveness and effectiveness choose a chart for data	Visual choice can be automated with explicit rules	Covers data charts only, not lecture text	**Extends rule-based choice to 27 slide families with confidence and reason**

Research gap

Earlier systems either generate new slides from a document, which changes the teacher's wording and can invent content, or select sentences without any visual design. Design research tells us what a good slide looks like but leaves the redesign to the teacher. No published system takes an existing lecture deck, keeps every point, explains each design decision and checks the AI output it adds. Learnova is built for exactly this gap.

Also reviewed

Tufte 2003 · Sweller 1988 · Cowan 2001 · Mayer 2009 · Knuth & Plass 1981 · Moritz 2019 · Ji 2023 · Kasneci 2023 · Renkl 2014 · Hundhausen 2002 · Wieman 2008 · Chi & Wylie 2014; tools: Gamma, Copilot, Gemini Slides, Canva. Full list with links on the next slide.

Learnova

RESEARCH AND REFERENCES

12 road signs · 36 sources · scan any code to check

THE PROBLEM
why old decks fail learners

- **Tufte 2003: Cognitive Style of PowerPoint**
  bullets fragment the argument
  edwardtufte.com
- **Sweller 1988: Cognitive load**
  working memory limits learning
  doi.org/10.1207/s15516709cog1202_4
- **Cowan 2001: the magical number 4**
  about four chunks at a time
  doi.org/10.1017/S0140525X01003922

SLIDE DESIGN
what a good slide looks like

- **Garner & Alley 2013, IJEE**
  assertion-evidence beats bullets
  ijee.ie
- **Alley et al. 2006**
  sentence headlines aid retention
  Technical Communication 53(2)
- **Mayer 2009: Multimedia Learning**
  coherence, signalling, contiguity
  cambridge.org

SLIDE GENERATION
prior systems we improve on

- **Fu et al. 2022: DOC2PPT**
  paper → slides, neural
  doi.org/10.1609/aaai.v36i1.19943
- **Sun et al. 2021: D2S**
  query-based summarisation
  aclanthology.org/2021.naacl-main.111
- **Hu & Wan 2015: PPSGen**
  learned sentence selection
  doi.org/10.1109/TKDE.2014.2359652

FAITHFULNESS
why we keep text verbatim

- **Maynez et al. 2020, ACL**
  summaries hallucinate and omit
  aclanthology.org/2020.acl-main.173
- **Ji et al. 2023, ACM CSUR**
  survey of hallucination
  doi.org/10.1145/3571730
- **Kasneci et al. 2023**
  LLMs in education: risks
  doi.org/10.1016/j.lindif.2023.102274

WORKED EXAMPLES
steps with reasons

- **Sweller & Cooper 1985**
  worked examples teach novices
  doi.org/10.1207/s1532690xci0201_3
- **Atkinson et al. 2000**
  principles of example study
  doi.org/10.3102/00346543070002181
- **Renkl 2014, Cognitive Science**
  example-based learning theory
  doi.org/10.1111/cogs.12086

INTERACTIVE LABS
learning by doing

- **Hundhausen et al. 2002**
  engagement drives visualisation gains
  doi.org/10.1006/jvlc.2002.0237
- **Wieman et al. 2008: PhET**
  simulations that teach
  doi.org/10.1126/science.1161948
- **Chi & Wylie 2014: ICAP**
  interactive > active > passive
  doi.org/10.1080/00461520.2014.965823

LAYOUT & DESIGN RULES
algorithms we adapted

- **Knuth & Plass 1981**
  optimal line breaking → CLASS
  doi.org/10.1002/spe.4380111102
- **Mackinlay 1986: APT**
  automatic visual design rules
  doi.org/10.1145/22949.22950
- **Moritz et al. 2019: Draco**
  design knowledge as constraints
  doi.org/10.1109/TVCG.2018.2865240

DOCUMENTS & OCR
reading real decks

- **Smith 2007: Tesseract OCR**
  open-source OCR engine
  github.com/tesseract-ocr
- **PyMuPDF + LibreOffice**
  PDF geometry, PPTX → PDF
  pymupdf.readthedocs.io
- **python-pptx**
  native, editable PowerPoint
  python-pptx.readthedocs.io

CODE AND TESTS
open, testable, honest

- **Source code**
  774 tests, all modules
  github.com/SDP42/LearnovaV2
- **Gold sets**
  101 labelled slides
  tests/fixtures on GitHub
- **Measurement scripts**
  every number on these slides
  scripts/research on GitHub

TRY IT
run it today

- **Run locally**
  uvicorn API + Vite web app
  README on GitHub
- **Demo decks**
  7 real decks rebuilt
  docs/review on GitHub
- **API docs**
  every endpoint, try it live
  localhost:8000/docs

TOOLS COMPARED
what exists today

- **Gamma**
  prompt → new deck
  gamma.app
- **Microsoft Copilot / Gemini Slides**
  AI inside office suites
  microsoft.com · workspace.google.com
- **Canva Magic Design**
  template-first AI slides
  canva.com

AI MODELS
what powers the AI layer

- **Google Gemini API**
  5-key pool, diagrams, titles
  ai.google.dev

Learnova

CONCLUSIONS OBTAINED

What the project set out to do, what we achieved, and what comes next

Objectives versus outcomes

Keep every point of the teacher's deck	98–100% of facts kept on 7 real decks
Give every slide a clear title	untitled slides 83 → 0 on 5 decks
Pick the right visual, with a reason	97% correct on 101 gold slides, < 1 ms each
Illustrate every topic slide	251 visual slides, 190 linked illustrations
Explain solved problems step by step	26 / 26 correct, 0 false alarms on 52 sentences
No overflow, no broken words	0 on every rebuilt deck
Work on any uploaded file	9 / 9 unseen files built; 2.3% slides flagged
Lower cognitive load on every deck	better on long PDFs, mixed on short decks

In one line

Learnova turns the decks teachers already have into faithful, visual and interactive lessons in about a minute, explains every design choice it makes, and uses AI only where its output can be checked.

Status: working prototype, tested on 7 real decks, ready for a department pilot.

Key conclusions

- **Redesign beats regeneration:** starting from the teacher's deck keeps it trusted and correct.
- **Rules + checked AI work together:** rules give speed and reasons; AI adds titles, diagrams and labs only after a check.
- **Exact maths matters:** SymPy solvers were right even where our own hand answers were wrong.
- **Measure honestly:** PSF showed where rebuilt slides were longer, and we report it.
- **Built for real classrooms:** free, offline, editable, and fast enough to use before a lecture.
- **Checks catch real bugs:** our own tests found and fixed a lab-gate and a key-pool fault.
- **Explainability builds trust:** teachers accept a visual faster when they can see why it was chosen.

Future scope

- Classroom study with our own students next semester
- Hindi and Marathi decks; voice narration
- LMS export (Moodle, Google Classroom)
- Calibrate PSF weights with student ratings
- Department server and shared lab library
- Handwritten-slide OCR and more solver types

Outputs: 2 research papers · IPD report · open-source code

Learnova

THANK YOU

LEARNOVA

Old decks. New learning.

A faithful, explainable and interactive redesign of the lecture decks teachers already have — every point kept, every slide illustrated, every AI output checked.

Team

Swayam Devendra Panchal · Kawaljeet Singh Bharaj
Jeevesh Bharat Bodhani · Rishit Pankajkumar Lahoti

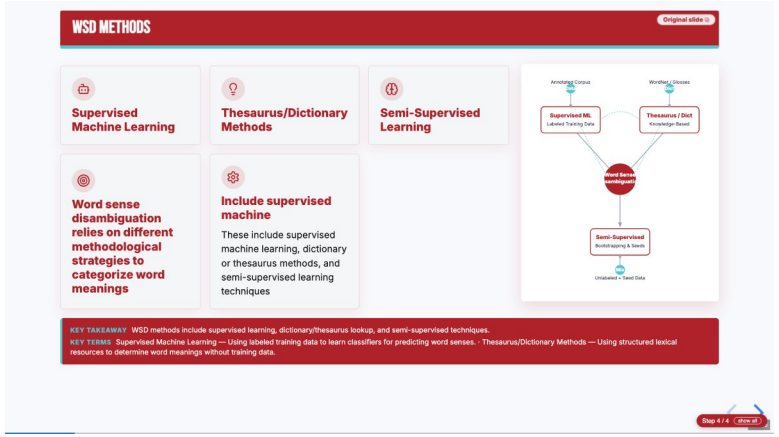
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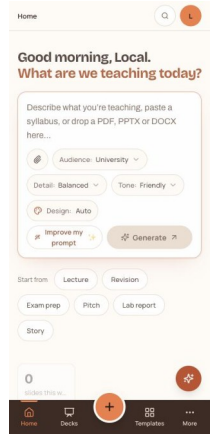
Dr. Pratik Kanani, Head of Department
Artificial Intelligence and Data Science, DJSCE, Mumbai

What it gives back

Editable PowerPoint · interactive web deck · study pack · labs · quizzes · step-by-step explainers · deck tutor

Questions and live demo welcome





83→0	98–100%	774	~35 s
untitled slides	facts kept	automated tests	216-page deck

Try it yourself

Scan for the source code, demo decks, test suite and both research papers.

github.com/SDP42/LearnovaV2

Upload any lecture deck: PPTX, PDF or Word. Get back an editable PowerPoint, an interactive web deck and a study pack.

Runs on one laptop: FastAPI + React, works offline, free.